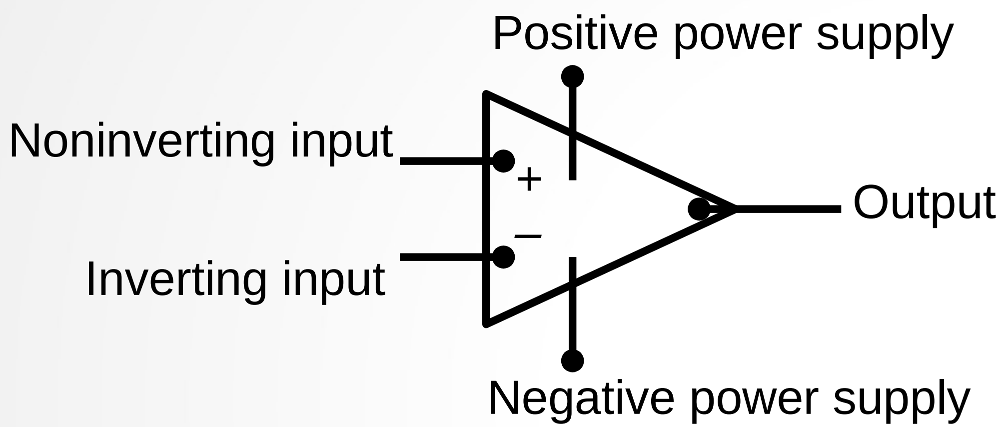
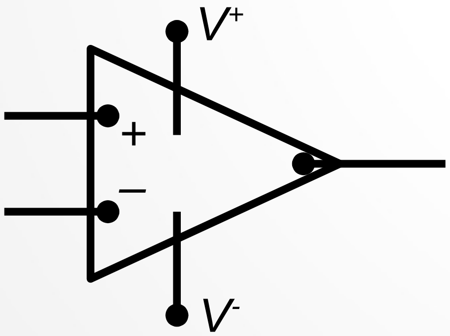
Chapter-5 The Operational Amplifier

It was referred to as operational because it was used to implement the mathematical operations of integration, differentiation, addition, sign changing, and scaling.

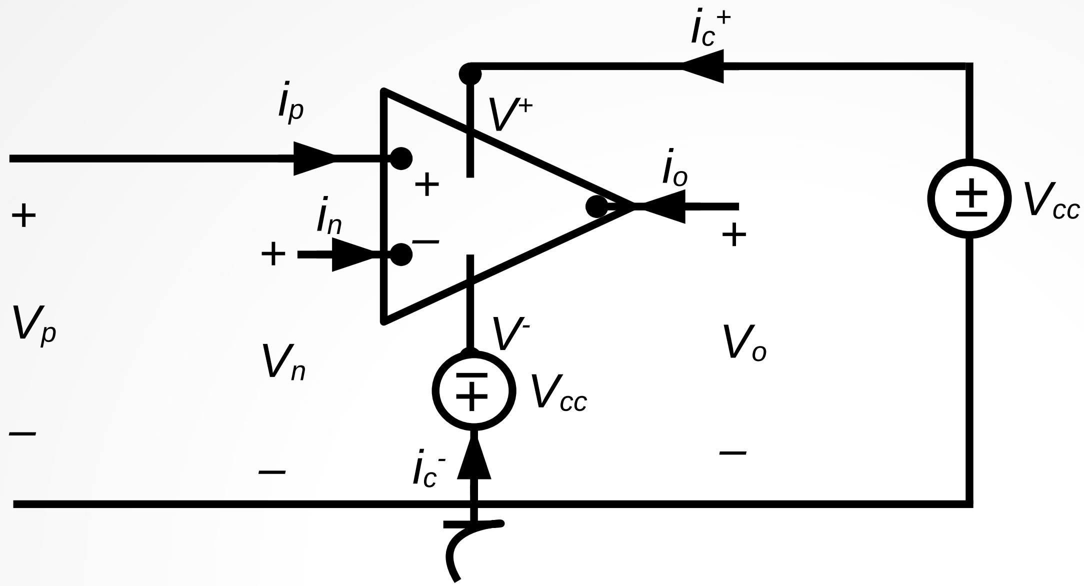


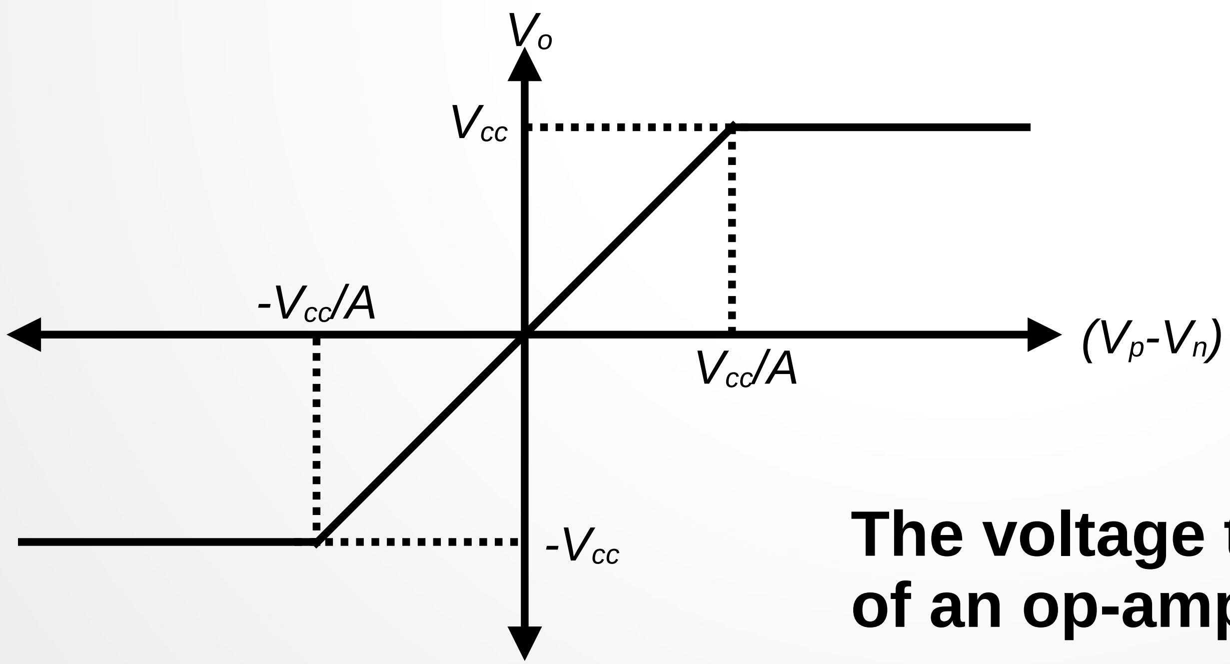
The circuit symbol for an operational amplifier (op-amp)



A simplified circuit symbol for an op-amp

Terminal current and voltage variables





The voltage transfer characteristic of an op-amp

Chapter-5 The Operational Amplifier

$$V_o = \begin{cases} -V_{cc} & A(V_p - V_n) < -V_{cc} \\ A(V_p - V_n) & -V_{cc} \leqslant A(V_p - V_n) \leqslant V_{cc} \quad \text{(linear region)} \\ V_{cc} & A(V_p - V_n) > V_{cc} \end{cases}$$

Constraints:

- $A > 10^4$ for the most of op-amps

$|V_{cc}| \leqslant 20\,v$ for the most of op-amps

$\Rightarrow$ For the linear region: $|V_p - V_n| < \dfrac{20}{10^4} = 2\,mv$

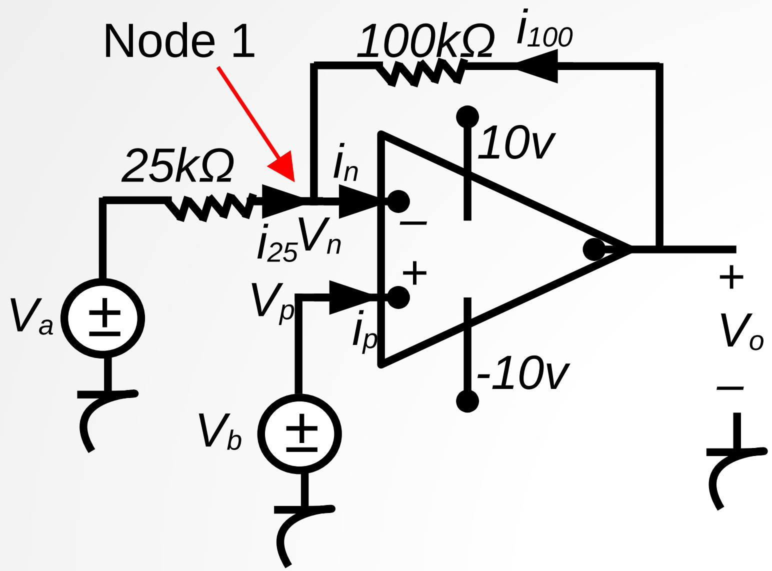
$\Rightarrow V_p \simeq V_n$

-The equivalent resistance seen by the input terminals of the op-amp is very large, typically *1MΩ* or more.

$\Rightarrow i_p = i_n \simeq 0$ Using KCL: $i_p + i_n + i_o + i_{c^+} + i_{c^-} = 0 \quad \Rightarrow i_o = -i_{c^+} - i_{c^-}$

For an ideal op-amp $A = \infty$, the input resistance $= \infty$

Example: The op-amp is ideal



a) V_o=? If V_a=1v and V_b=0v
b) V_o=? If V_a=1v and V_b=2v
c) If V_a=1.5v, specify the range of V_b that avoids amplifier saturation.

Since the op-amp is ideal, $i_p = i_n = 0, \quad V_p = V_n = V_b$

KCL for node-1

$$i_{25} + i_{100} = i_n = 0 \quad \Rightarrow \frac{V_a - V_n}{25000} + \frac{V_o - V_n}{100000} = 0 \Rightarrow \frac{V_a - V_b}{25000} + \frac{V_o - V_b}{100000} = 0$$

$\Rightarrow V_o = 5V_b - 4V_a$

a) $V_o = -4v$

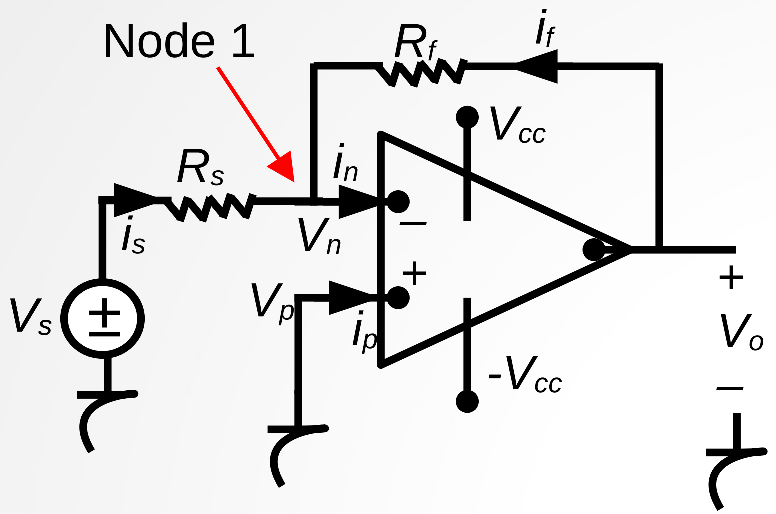
b) $V_o = 5*2 - 4*1 = 6v$

c) $-10 \leqslant V_o \leqslant 10$ to work in the linear region $\Rightarrow -10 \leqslant 5V_b - 4V_a \leqslant 10$

$\Rightarrow -10 \leqslant 5V_b - 6 \leqslant 10 \quad \Rightarrow -4 \leqslant 5V_b \leqslant 16 \quad \Rightarrow -0.8v \leqslant V_b \leqslant 3.2v$

If $-0.8v \leqslant V_b \leqslant 3.2v$, the op-amp will work in the linear region

The Invertering-Amplifier Circuit



R_s: source resistor

R_f: feedback resistor

The op-amp is ideal, $\Rightarrow i_p = i_n = 0, \quad V_p = V_n = 0$

KCL for node-1

$$i_s + i_f = i_n = 0 \quad \Rightarrow \frac{V_s - V_n}{R_s} + \frac{V_o - V_n}{R_f} = 0 \Rightarrow \frac{V_s}{R_s} + \frac{V_o}{R_f} = 0$$

$$\Rightarrow V_o = -\frac{R_f}{R_s} V_s$$

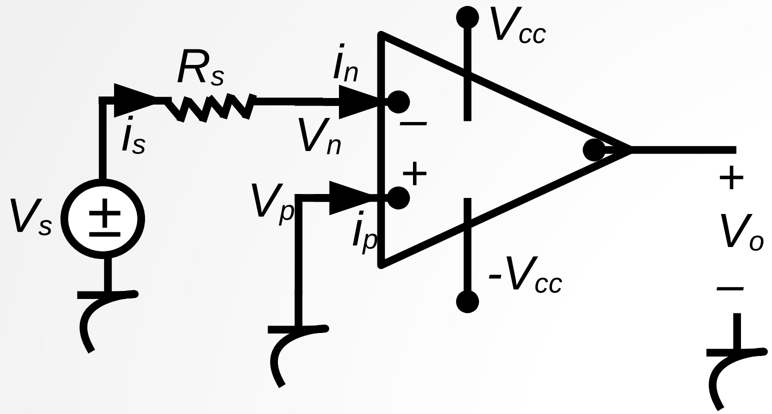
$\frac{R_f}{R_s}$: Gain

If we want the op-amp to work in the linear region

$$|V_o| \leqslant V_{cc} \quad \Rightarrow \left| \frac{R_f}{R_s} V_s \right| \leqslant V_{cc} \quad \Rightarrow \frac{R_f}{R_s} \leqslant \left| \frac{V_{cc}}{V_s} \right|$$

For example if $V_{cc} = 15\,v$ and maximum $V_s = 10\,mv$, $\frac{R_f}{R_s}$ must be less than 1500 to work in the linear region.

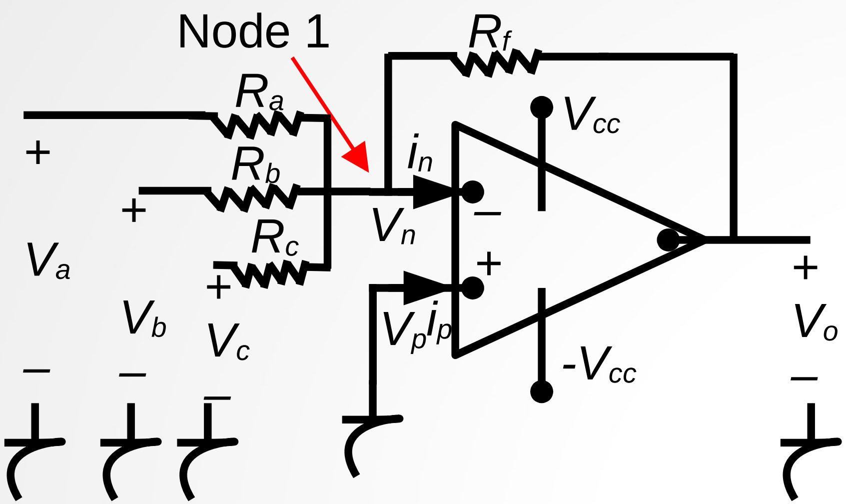
Openining the feedback



$$\Rightarrow V_o = -A * V_n \simeq -A * V_s$$

An inverting amplifier operating open loop

The Summing-Amplifier Circuit



$$\text{If } R_a = R_b = R_c = R_s$$

$$\Rightarrow V_o = -\frac{R_f}{R_s}\left(V_a + V_b + V_c\right)$$

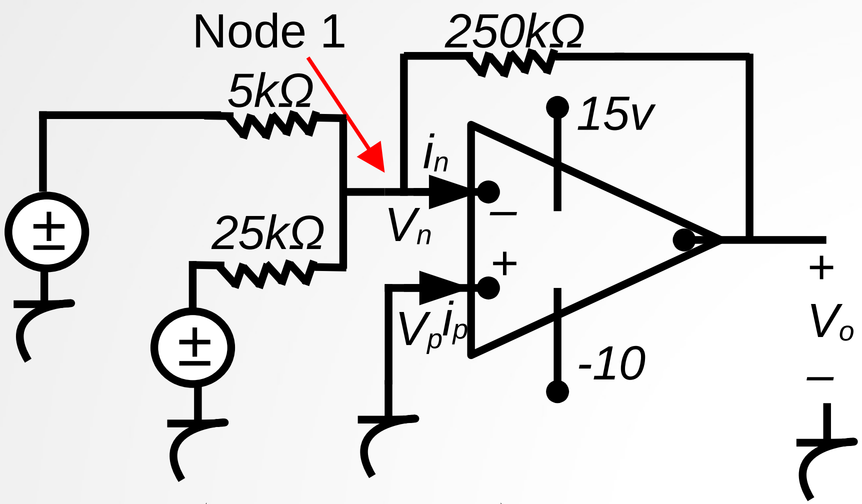
The op-amp is ideal, $\Rightarrow i_p = i_n = 0, \quad V_p = V_n = 0$

KCL for node-1

$$\frac{V_n - V_a}{R_a} + \frac{V_n - V_b}{R_b} + \frac{V_n - V_c}{R_c} + \frac{V_n - V_o}{R_f} + i_n = 0$$

$$\Rightarrow V_o = -\left(\frac{R_f}{R_a} V_a + \frac{R_f}{R_b} V_b + \frac{R_f}{R_c} V_c\right)$$

Example



a) V_o=? If V_a=0.1v and V_b=0.25v
b) If V_b=0.25v, how large can V_a be before the -op-amp saturates.
c) If V_a=0.1, how large can V_b be before the -op-amp saturates.
d) Repeat a,b,c with the polarity of V_b reversed.

a) $V_o=-\left(\frac{250}{5}V_a+\frac{250}{25}\right)=-50V_a-10V_b$

$\Rightarrow V_o=-50*0.1-10*0.25=-7.5v$

b) $V_o=-50*V_a-10*0.25=-10 \quad \Rightarrow -50V_a=-7.5$

$\Rightarrow V_a=0.15v$ The op-amp will saturate if $V_a\geq 0.15v$

c) $V_o=-50*0.1-10*V_b=-10 \quad \Rightarrow -10V_b=-5$

$\Rightarrow V_b=0.5v$ The op-amp will saturate if $V_b\geq 0.5v$

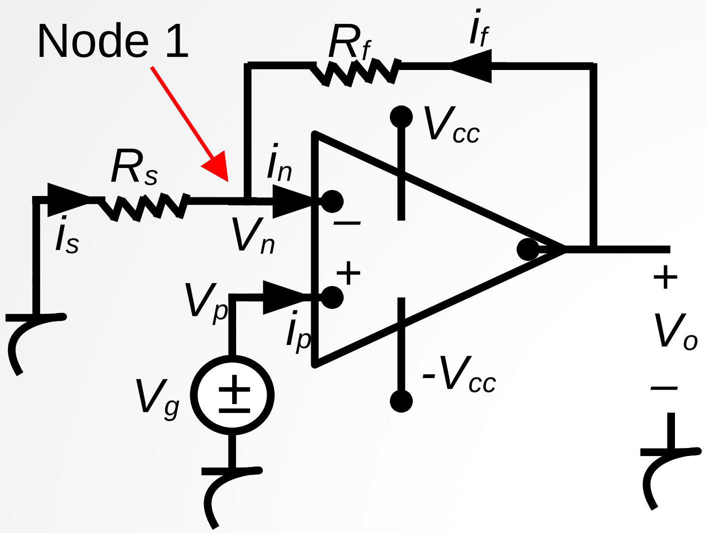
d) i) $V_o=-50*V_a+10*V_b=-50*0.1+10*0.25=-2.5v$

ii) $V_o=-50*V_a+10*0.25=-10 \quad \Rightarrow -50V_a=-12.5$

$\Rightarrow V_a=0.25v$ The op-amp will saturate if $V_a\geq 0.25v$

iii) $V_o=-50*0.1+10*V_b=15 \quad \Rightarrow 10V_b=20 \Rightarrow V_b=2v$ The op-amp will saturate if $V_b\geq 2v$

The Noninvertering-Amplifier Circuit



The op-amp is ideal, $\Rightarrow i_p = i_n = 0, \quad V_n = V_p = V_g$

KCL for node-1

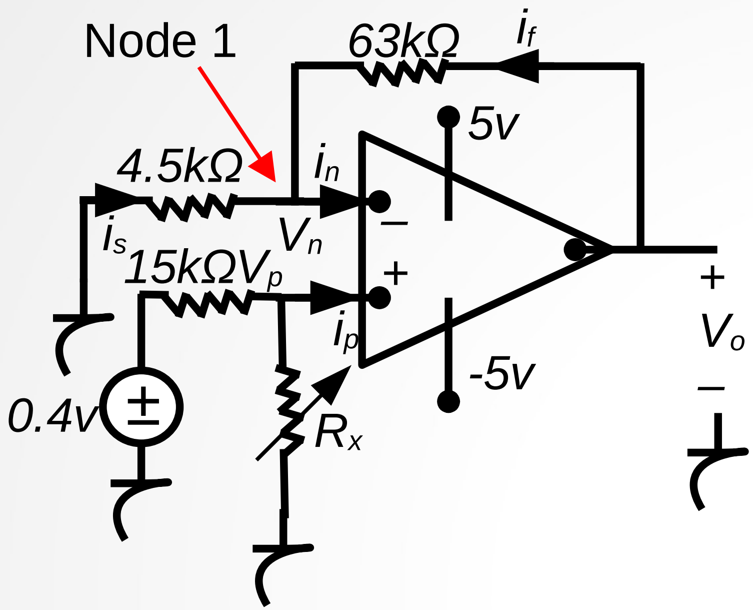
$$\frac{V_n}{R_s} + \frac{V_n - V_o}{R_f} = 0 \Rightarrow \frac{V_g}{R_s} + \frac{V_g}{R_f} = \frac{V_o}{R_f}$$

$$\Rightarrow V_o = V_g \frac{R_s + R_f}{R_s}$$

Operating in the linear region requires taht

$$\frac{R_s + R_f}{R_s} < \left| \frac{V_{cc}}{V_g} \right|$$

Example



a) Find the output voltage when the variable resistor is set to 60 kΩ.

b) how large Rx can be before the amplifier saturates?

The op-amp is ideal, $\Rightarrow i_p = i_n = 0, \quad V_n = V_p = V_1$

$$V_p = 0.4 * \frac{60\,k\Omega}{60\,k\Omega + 15\,k\Omega} = 0.32\,v = V_1$$

a)

KCL for node-1

$$\frac{V_1}{4.5\,k\Omega} + \frac{V_1 - V_o}{63\,k\Omega} = 0 \Rightarrow V_o = \frac{67.5}{4.5} * V_1 = 4.8\,v$$

b)

$$V_p = 0.4 * \frac{R_x}{R_x + 15\,k\Omega} = V_1$$

KCL for node-1

$$\frac{V_1}{4.5\,k\Omega} + \frac{V_1 - V_o}{63\,k\Omega} = 0$$

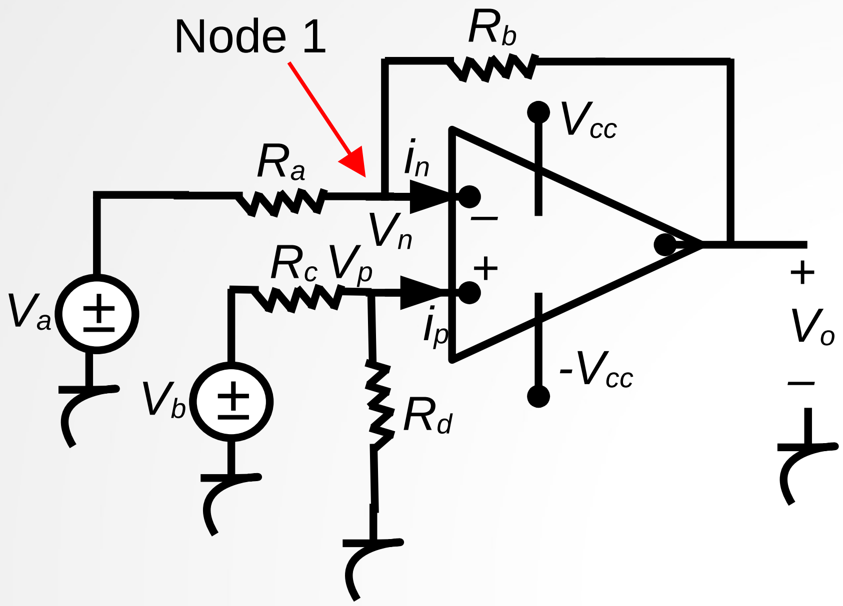
$$\Rightarrow V_o = \frac{67.5}{4.5} * V_1 = \frac{67.5}{4.5} * 0.4 * \frac{R_x}{R_x + 15\,k\Omega}$$

The op-amp will saturate when V_o reaches to $5\,v$.

$$\Rightarrow V_o = 6 * \frac{R_x}{R_x + 15\,k\Omega} = 5\,v$$

$$\Rightarrow R_x = 5 * 15\,k\Omega = 75\,k\Omega$$

The Difference Amplifier Circuit



The op-amp is ideal, $\Rightarrow i_p = i_n = 0, \quad V_n = V_p = V_1$

$$V_p = \frac{R_d}{R_d + R_c} V_b = V_1$$

a)

KCL for node-1

$$\frac{V_1 - V_a}{R_a} + \frac{V_1 - V_o}{R_b} = 0$$

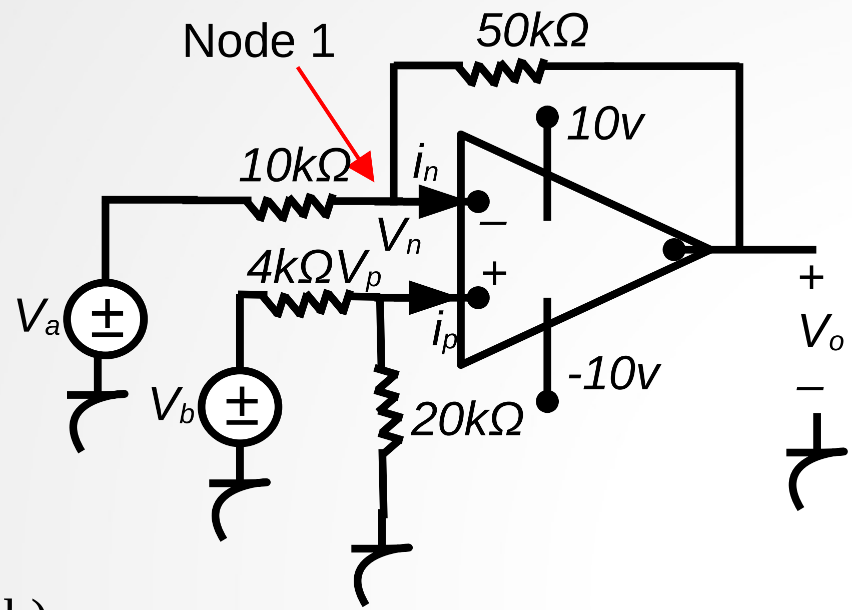
$$\Rightarrow V_o = V_1 * \left(\frac{R_b}{R_a} + 1\right) - V_a * \frac{R_b}{R_a}$$

$$\Rightarrow V_o = \frac{R_d (R_a + R_b)}{R_a (R_c + R_d)} * V_b - \frac{R_b}{R_a} * V_a$$

If $\dfrac{R_a}{R_b} = \dfrac{R_c}{R_d}$

$$V_o = \frac{R_b}{R_a} * (V_b - V_a)$$

Example



a) What range of values for V_a will result in linear operation if V_b=4v?
b) Repeat (a) with 20kΩ resistor decreased to 8kΩ.

The op-amp is ideal, $\Rightarrow i_p = i_n = 0$,

$V_n = V_p = V_1$

$$V_p = \frac{20}{20+4} V_b = \frac{5}{6} V_b = V_1$$

a)

KCL for node-1

$$\frac{V_1 - V_a}{10k\Omega} + \frac{V_1 - V_o}{50k\Omega} = 0 \quad \Rightarrow V_o = 6V_1 - 5V_a$$

$\Rightarrow V_o = 5*(V_b - V_a)$

If $V_b = 4v \quad \Rightarrow V_o = 5*(4 - V_a)$

$-10 \le V_o \le 10 \quad \Rightarrow -10 \le 5*(4 - V_a) \le 10$

$\Rightarrow 2 \le V_a \le 6$

The op-amp will operate in the linear region if $2 \le V_a \le 6$.

b)

$$V_p = \frac{8}{8+4} V_b = V_1$$

$V_o = 6V_1 - 5V_a = 4V_b - 5V_a$

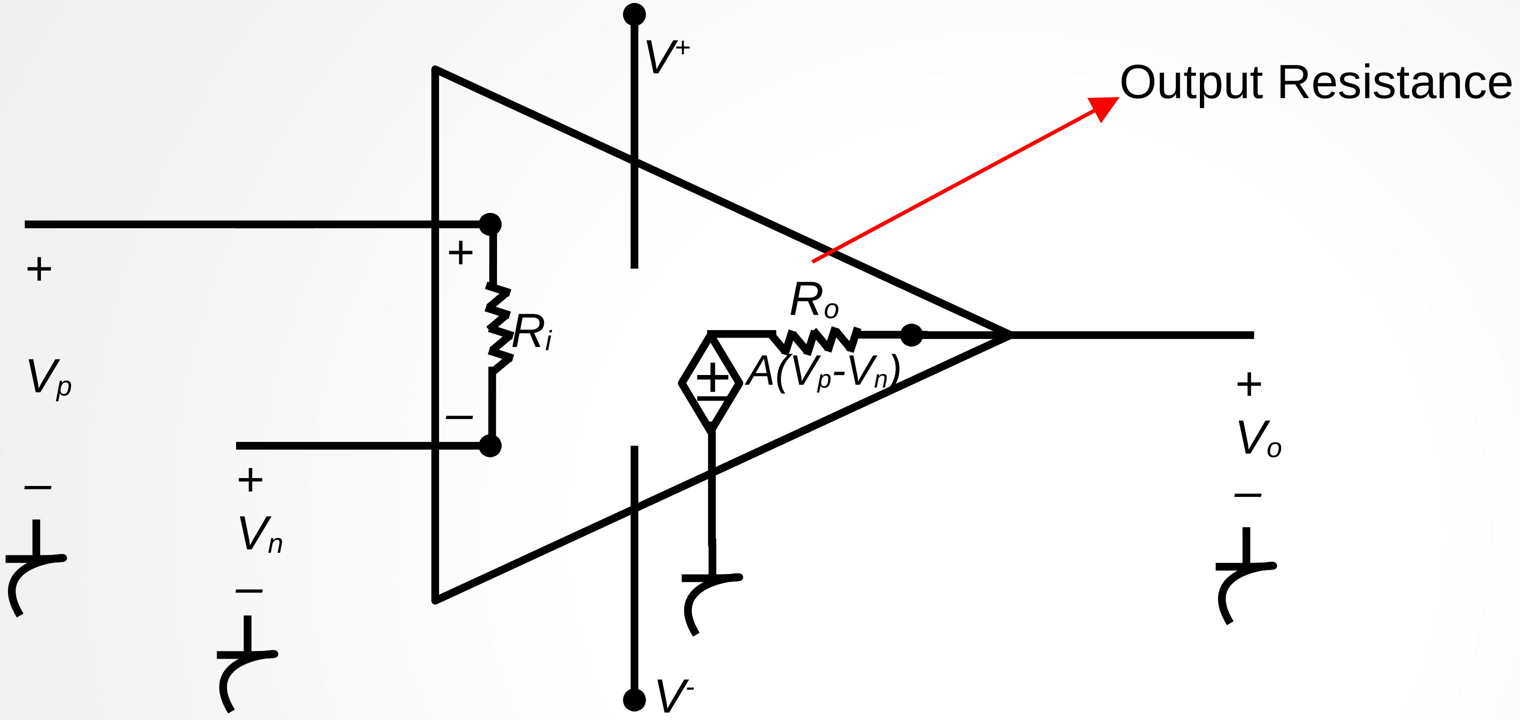
If $V_b = 4v \quad \Rightarrow V_o = 16 - 5V_a$¿

$-10 \le V_o \le 10 \quad \Rightarrow -10 \le 16 - 5V_a \le 10$

$\Rightarrow 1.2 \le V_a \le 5.2$

The op-amp will operate in the linear region if $1.2 \le V_a \le 5.2$.

A More Realistic Model for the Operational Amplifier



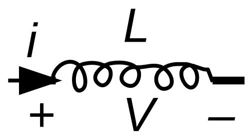
Chapter-6 Inductance, Capacitance andMutual Inductance

The Inductor

-An inductor is a passive element designerd to store energy in its magnetic field. They are used in power supplies, transformers, radios, TV's, radars, and electric motors.

-An inductor is composed of a coil of wire wound around a supporting core whose material may be magnetic or nonmagnetic.

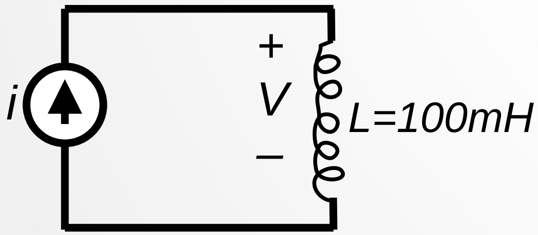
-If the current is varying with time, the magnetic field is varying with time. A time varying magnetic field induces a voltage in any conductor linked by the field.



Inductance in Henry

$$V = L\frac{di}{dt}$$

Example



$$i = 0\,A \quad \text{for} \quad t \leq 0$$
$$i = 10\,te^{-5t} \quad \text{for} \quad t \geq 0$$
$$V(t) = ?$$

$$V(t) = L\frac{di(t)}{dt}$$

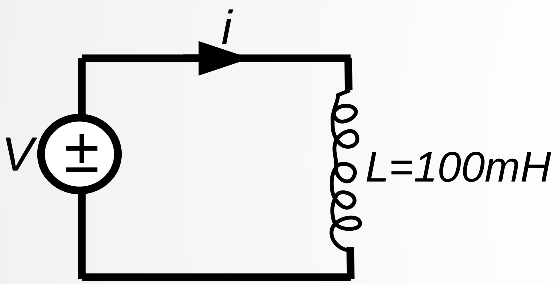
$$= 0.1 * \frac{10\,te^{-5t}}{dt} = e^{-5t} - 5t\,e^{-5t} = e^{-5t} * (1 - 5t) \quad t \geq 0$$

Current in an inductor in term of the voltage across the inductor

$$V=L\frac{di}{dt} \quad \Rightarrow Vdt=L\frac{di}{dt}dt \quad \Rightarrow Vdt=L\,di$$

$$\Rightarrow L\int_{i(t_0)}^{i(t)} dx=\int_{t_0}^{t} V(\tau)d\tau \Rightarrow i(t)=\frac{1}{L}\int_{t_0}^{t} V(\tau)d\tau+i(t_0)$$

Example



$V=0\,v \quad$ for $\quad t\le 0$

$V=20te^{-10t} \quad$ for $\quad t\ge 0$

Also assume $i=0$ for $t\le 0$

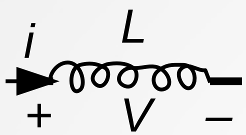
$i(t)=?$

$$i(t)=\frac{1}{L}\int_{t_0}^{t} V(\tau)d\tau+i(t_0)=\frac{1}{0.1}\int_{0}^{t} 20\tau e^{-10\tau}d\tau+0$$

$$\Rightarrow i(t)=200\left[-\frac{e^{-10\tau}}{100}*(100\tau+1)\right]_0^t$$

$$=2\left(1-10te^{-10t}-e^{-10t}\right)A \quad t\ge 0$$

Power and energy in the Inductor



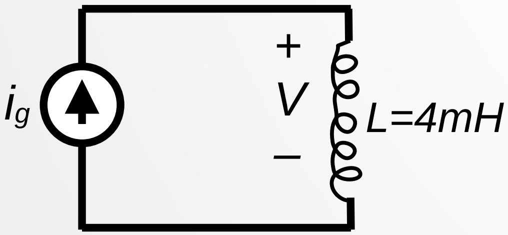
$$P=V*i=Li\frac{di}{dt}=V*\left[\frac{1}{L}\int_{t_0}^{t}V(\tau)d\tau+i(t_0)\right]$$

W: Energy

$$P=\frac{dW}{dt}=Li\frac{di}{dt}\quad\Rightarrow dW=L*idi$$

$$\Rightarrow\int_{0}^{W}dx=L\int_{0}^{i}ydy\quad\Rightarrow W=\frac{1}{2}L*i^2$$

Example



$i_g(t)=0\,A$ for $t\leq 0$

$i_g(t)=8e^{-300t}-8e^{-1200t}\,A$ for $t\geq 0$

a) $V(0)=?$

b) The instant of time, greater than zero, when the voltage V passes through zero?

c) $P_V(t)=?$

d) The instant when the power delivered to the inductor is maximum?

e) The maximum power for the inductor?

f) The maximum energy stored in the inductor?

g) The instant of time when the stored energy is maximum?

Solution

a) $V(t)=Li\dfrac{di}{dt}=4*10^{-3}\left[9600\,e^{-1200\,t}-2400\,e^{-300\,t}\right] \quad \Rightarrow V(0)=4*(9.6-2.4)=28.8\,v$

b) $V(t)=0=4*10^{-3}\left[9600\,e^{-1200\,t}-2400\,e^{-300\,t}\right]$

$\Rightarrow 9.6*e^{-1200\,t}=2.4*e^{-300\,t} \quad \Rightarrow -900\,t=\ln\dfrac{2.4}{9.6} \quad \Rightarrow t=1.54\,msec\,.$

c) $P(t)=V(t)*i_g(t)=4*\left[9.6\,e^{-1200\,t}-2.4\,e^{-300t}\right]*8*\left(e^{-300\,t}-8\,e^{-1200\,t}\right)A \quad \text{for} \quad t\geq 0$

$\quad =76.8*e^{-600\,t}+384*e^{-1500\,t}-307.2*e^{-2400\,t}$

d) $\dfrac{dP(t)}{dt}=0=e^{-600\,t}\left[46080-576000*e^{-900\,t}+737280*e^{-1800\,t}\right]=0$

$\Rightarrow 46080-576000*y+737280*y^2=0 \quad \Rightarrow t=411.05\,\mu\,sec\,.$

e) $P_{max}=P(411.05\,\mu\,sec)=32.72\,w$

f) $W=\dfrac{1}{2}L*i^2 \quad \Rightarrow W_{max}=\dfrac{1}{2}L*i_{max}^2=\dfrac{1}{2}*4*10^{-3}*i^2(1.54\,msec)$

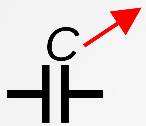
$\quad =2*4*10^{-3}*64*[e^{-0.3*1.54}-e^{-1.2*1.54}\text{¿}]^2=28.57\,mJ\,.$

g) The instant of time $=1.54$ msec. (the energy will be maximum when the current is maximum)

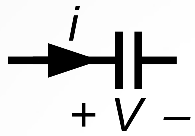
Chapter-6 Inductance, Capacitance andMutual Inductance

The Capacitor

-A capacitor is an electrical component that consists of two conductors seperated by an insulator or dielectric material.



Capacitance in Farad



$C:$ capacitance of the capacitor $\quad q:$ charge

$$q = C * V$$

$$i = \frac{dq}{dt} = C * \frac{dV}{dt}$$

$$idt = CdV \Rightarrow \int_{V(t_0)}^{V(t)} dx = \frac{1}{C}\int_{t_0}^{t} i(\tau)\,d\tau \Rightarrow V(t) = \frac{1}{C}\int_{t_0}^{t} i(\tau)\,d\tau + V(t_0)$$

$$\text{if } t_0 = 0 \quad \Rightarrow V(t) = \frac{1}{C}\int_{0}^{t} i(\tau)\,d\tau + V(0)$$

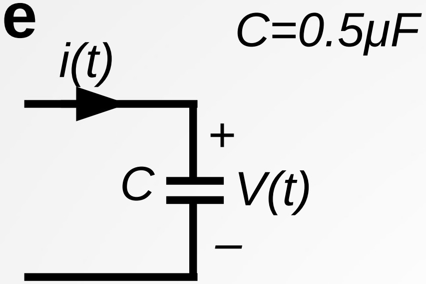
$$P = V * i = CV\frac{dV}{dt} = i * \left[\frac{1}{C}\int_{t_0}^{t} i(\tau)\,d\tau + V(t_0)\right]$$

W: Energy

$$P = \frac{dW}{dt} = CV\frac{dV}{dt} \quad \Rightarrow dW = C * VdV$$

$$\Rightarrow \int_{0}^{W} dx = C\int_{0}^{V} ydy \quad \Rightarrow W = \frac{1}{2}C * V^2$$

Example



$$V(t)=\begin{cases} 0 & t<0 \\ 4t & 0\le t\le 1\,sec \\ 4e^{-(t-1)} & t\ge 1\,sec \end{cases}$$

a) i(t)=?, P(t)=?,   W(t)=?
b) Specify the interval of time when energy is being stored in the capacitor.
c) Specify the interval of time when energy is being delivered by the capacitor.
Solution

$$i(t)=C\frac{dV(t)}{dt}=\begin{cases} 0 & t\le 0 \\ 2\mu A & 0\le t\le 1\,sec \\ -2e^{-(t-1)}\mu A & t\ge 1\,sec \end{cases}$$

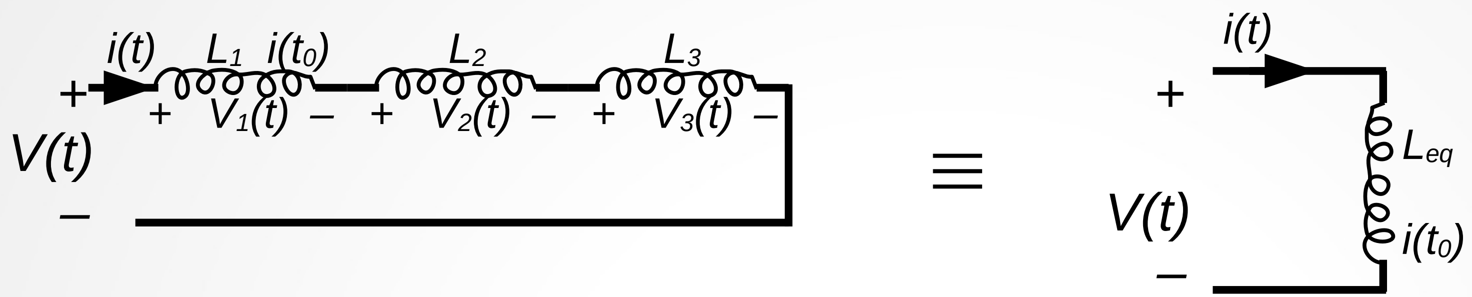
$$P(t)=i(t)V(t)=\begin{cases} 0 & t\le 0 \\ 8t\,\mu W & 0\le t\le 1\,sec \\ -8e^{-2(t-1)}\mu W & t\ge 1\,sec \end{cases}$$

$$W(t)=\frac{1}{2}CV^2(t)=\begin{cases} 0 & t\leq 0 \\ 4t^2 \mu J & 0\leq t\leq 1\,sec \\ 4e^{-2(t-1)}\mu J & t\geq 1\,sec \end{cases}$$

b) The energy is stored if the power is positive. The the energy is being stored in the interval 0-1sec.

c) The energy is being delivered by the capacitor whenever the power is negative. The energ is being delivered for all *t* greater than 1 sec.

Series-Parallel Combinations of Inductance and Capacitance
-Inductors in Series:



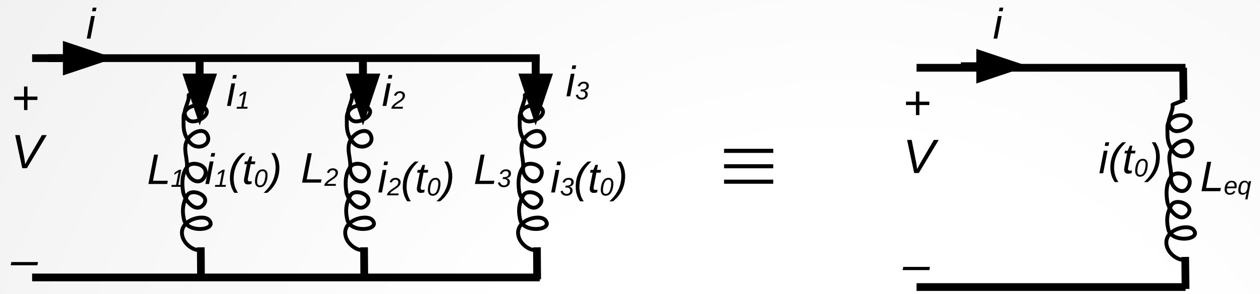
$$V(t) = L_{eq}\frac{di(t)}{dt}$$

$$V_1(t) = L_1\frac{di(t)}{dt}, \quad V_2 = L_2\frac{di(t)}{dt}, \quad V_3 = L_3\frac{di(t)}{dt}$$

$$V(t) = V_1(t) + V_2(t) + V_3(t) = (L_1 + L_2 + L_3)\frac{di(t)}{dt} = L_{eq}\frac{di(t)}{dt}$$

$$\Rightarrow L_{eq} = L_1 + L_2 + L_3$$

Combining Inductors in Parallel



$$i(t)=\frac{1}{L_{eq}}\int_{t_0}^{t} V(\tau)d\tau+i(t_0)$$

$$i_1(t)=\frac{1}{L_1}\int_{t_0}^{t} V(\tau)d\tau+i_1(t_0)$$

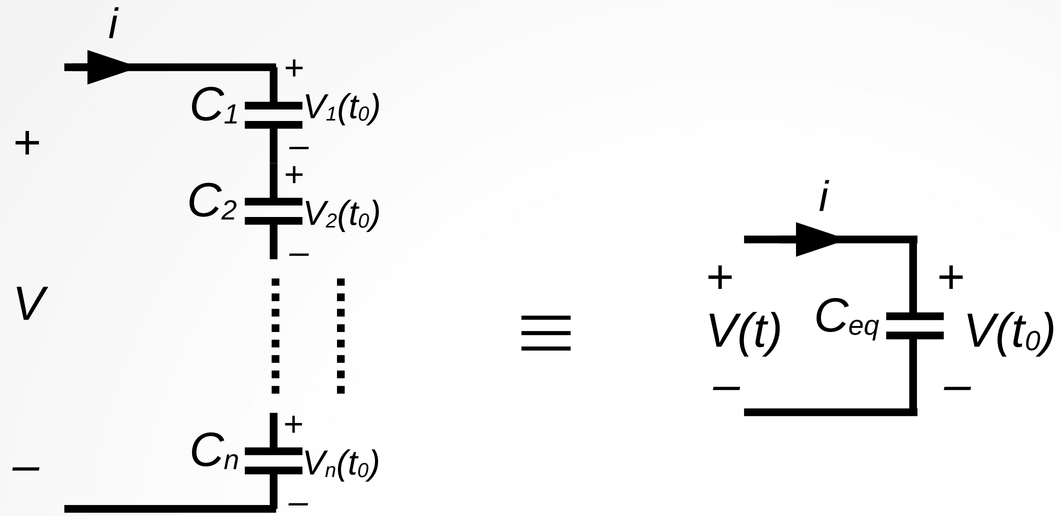
$$i_2(t)=\frac{1}{L_2}\int_{t_0}^{t} V(\tau)d\tau+i_2(t_0)$$

$$i_3(t)=\frac{1}{L_3}\int_{t_0}^{t} V(\tau)d\tau+i_3(t_0)$$

$$i(t)=i_1(t)+i_2(t)+i_3(t)=\left(\frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{L_3}\right)\int_{t_0}^{t} V(\tau)d\tau+i_1(t_0)+i_2(t_0)+i_3(t_0)=\frac{1}{L_{eq}}\int_{t_0}^{t} V(\tau)d\tau+i(t_0)$$

$$\Rightarrow\frac{1}{L_{eq}}=\left(\frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{L_3}\right),\qquad i(t_0)=i_1(t_0)+i_2(t_0)+i_3(t_0)$$

Capacitors Connected in Series



$$V(t)=\frac{1}{C_{eq}}\int_{t_0}^{t} i(\tau)d\tau+V(t_0)$$

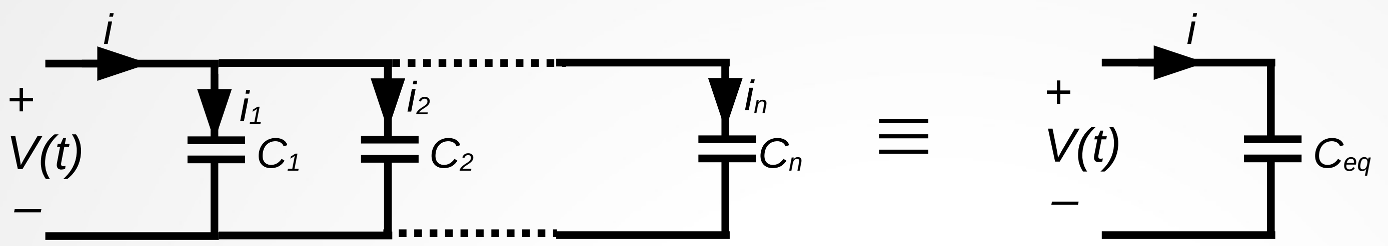
$$V(t)=V_1+V_2+\cdots+V_n=\frac{1}{C_1}\int_{t_0}^{t} i(\tau)d\tau+V_1(t_0)+\frac{1}{C_2}\int_{t_0}^{t} i(\tau)d\tau+V_2(t_0)+\cdots+\frac{1}{C_n}\int_{t_0}^{t} i(\tau)d\tau+V_n(t_0)$$

$$=\left(\frac{1}{C_1}+\frac{1}{C_2}+\cdots+\frac{1}{C_n}\right)\int_{t_0}^{t} i(\tau)d\tau+V_1(t_0)+V_2(t_0)+\cdots+V_n(t_0)$$

$$\Rightarrow\frac{1}{C_{eq}}=\left(\frac{1}{C_1}+\frac{1}{C_2}+\cdots+\frac{1}{C_n}\right),$$

$$V(t_0)=V_1(t_0)+V_2(t_0)+\cdots+V_n(t_0)$$

Capacitors Connected in Parallel

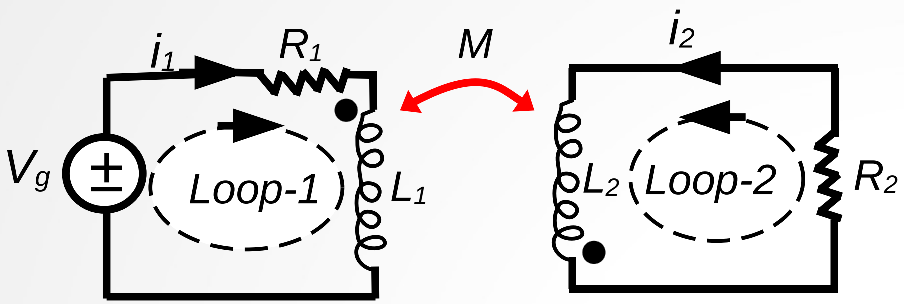


$$i(t)=C_{eq}\frac{dV(t)}{dt}$$

$$i(t)=i_1(t)+i_2(t)+\cdots+i_n(t)=C_1\frac{dV(t)}{dt}+C_2\frac{dV(t)}{dt}+\cdots+C_n\frac{dV(t)}{dt}=(C_1+C_2+\cdots+C_n)\frac{dV(t)}{dt}$$

$$C_{eq}=C_1+C_2+\cdots+C_n$$

Mutual Inductance



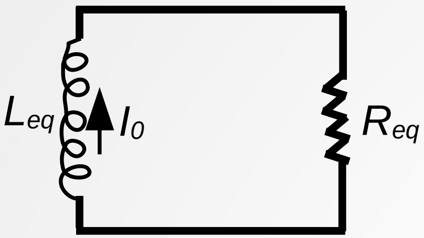
KVL for loop-1:

$$-V_g + i_1 R_1 + L_1 \frac{di_1}{dt} - M \frac{di_2}{dt} = 0$$

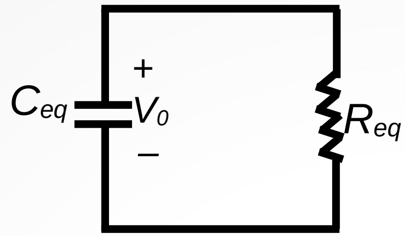
KVL for loop-2:

$$i_2 R_2 + L_2 \frac{di_2}{dt} - M \frac{di_1}{dt} = 0$$

The Natural Response of an RL and RC Circuits



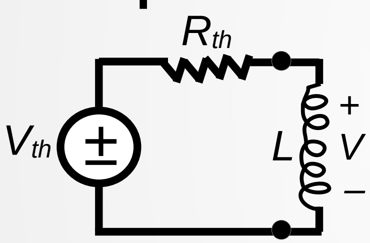
a)



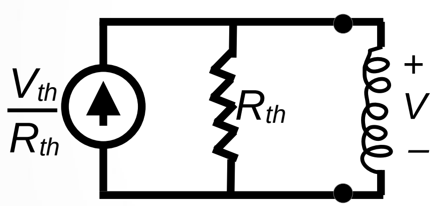
b)

Two forms of the circuit for natural response. a) RL circuit. b) RC circuit

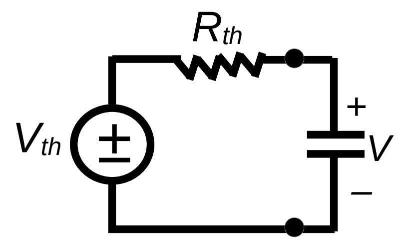
Four possible first-order circuits



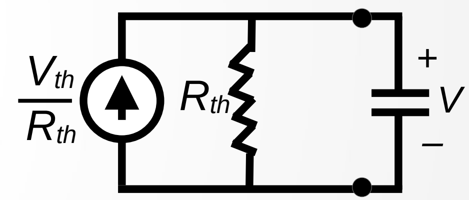
a) An inductor connected To a Thevenin equivalent



b) An inductor connected to a Norton equivalent

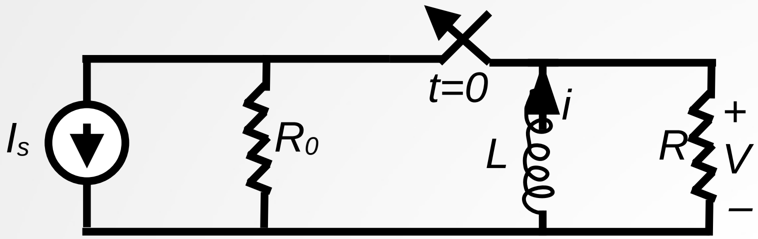


c) A capacitor connected to a Thevenin equivalent



d) A capacitor connected to a Norton equivalent

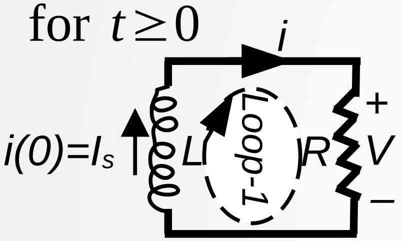
The Natural Response of an RL Circuit



The switch is opened at time *t=0*.
(The switch is disconnected at time *t=0*)

$i(0^-)=I_s$

for $t \geq 0$



$i(0^-)=i(0^+)=I_s$ Inductor current can not be changed immediately

KVL for loop-1

$$L\frac{di}{dt}+Ri=0 \quad \Rightarrow \frac{di}{i}=-\frac{R}{L}dt \quad \Rightarrow \int_{i(t_0)}^{i(t)}\frac{dx}{x}=-\frac{R}{L}\int_{t_0}^{t}dy$$

$$\Rightarrow \ln\frac{i(t)}{i(t_0)}=-\frac{R}{L}(t-t_0) \quad \Rightarrow i(t)=i(t_0)e^{-\frac{R}{L}(t-t_0)} \quad \text{for } t \geq t_0 \quad \text{If } t_0=0 \quad i(t)=i(0)e^{-\frac{R}{L}t} \quad \text{for } t \geq 0$$

$$V=Ri=I_sRe^{-\frac{R}{L}(t)} \quad t \geq 0^+ \qquad V(0^-)=0 \qquad V=L\frac{di}{dt}=0 \quad \text{for } t \leq 0^-$$

Chapter-7 Response of First Order RL and RC Circuits

The power dissipated in the resistor:

$$P = Vi = i^2 R = I_s^2 R e^{-2\frac{R}{L}t} \qquad t \geq 0^+$$

The energy delivered to the resistor during interval of time after the switch hase been opened is

$$W = \int_0^t P(x)\,dx = \int_0^t I_s^2 R e^{-2\frac{R}{L}x} dx = \frac{LRI_s^2}{2R}\left(1 - e^{-2\frac{R}{L}t}\right) = \frac{1}{2} LI_s^2\left(1 - e^{-2\frac{R}{L}t}\right) \qquad t \geq 0^+$$

The significance of the time constant

$$\tau = \text{ time constant } = \frac{L}{R} \text{ time constant for the } RL \text{ circuit}$$

$$i(t) = I_s e^{-\frac{R}{L}t} = I_s e^{-\frac{t}{\tau}} \quad \text{for } t \geq 0$$

$$V(t) = I_s R e^{-\frac{t}{\tau}} \quad \text{for } t \geq 0^+$$

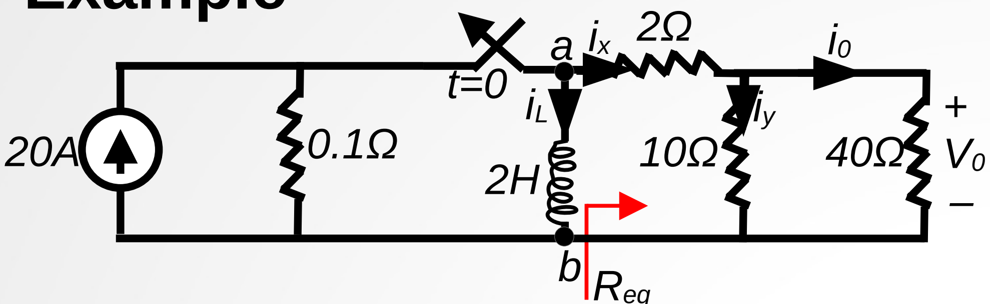
$$P = I_s^2 R e^{-2\frac{t}{\tau}} \quad \text{for } t \geq 0^+$$

$$W = \frac{1}{2} L I_s^2 \left(1 - e^{-2\frac{t}{\tau}}\right) \quad \text{for } t \geq 0^+$$

$$\text{If } t = 5\tau \quad \Rightarrow i(t) \leq \frac{1}{100} i(0) \text{ and } V(t) \leq \frac{1}{100} V(0^+)$$

-Thus, we sometimes say that, five time constants after switching, the current and voltages have, for most practical purposes, reached their final values.

Example



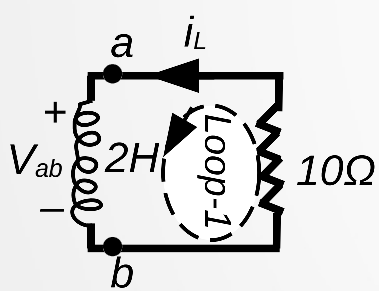
-The switch has been closed for a long time before it is opened at time *t=0*.
-Find **a)** $i_L(t)$ for t≥0 **b)** $i_0(t)$ for t≥0$^+$ **c)** $V_0(t)$ for t≥0$^+$ **d)** The percentage of the total energy stored in 2H inductor that is dissipated in the 10 Ω resistor.

Solution:

$$i_L(0^-)=i_L(0^+)=20\,A$$

$$R_{eq}=2+(40\,\Omega\,//\,10\,\Omega)=2+\frac{40*10}{40+10}=10\,\Omega$$

for $t\geq 0^+$



KVL for loop-1 $\quad L\frac{di_L}{dt}+Ri_L=0$

$$\Rightarrow i_L(t)=i_L(0)e^{-\frac{R}{L}t}=20\,e^{-\frac{10}{2}t}=20\,e^{-5t}\quad t\geq 0$$

$$V_{ab}=L\frac{di_L}{dt}=-Ri_L=-200\,e^{-5t}\quad t\geq 0^+$$

$$i_x=\frac{V_{ab}}{10}=-20\,e^{-5t}\,A\quad t\geq 0^+$$

$$i_0(t)=\frac{10}{10+40}i_x=-4\,e^{-5t}\,A\quad t\geq 0^+$$

c) $V_0(t)=40*i_0(t)=-160\,e^{-5t}\,V\quad t\geq 0^+$

The total energy stored in L at time 0:

$$W_L(0)=\frac{1}{2}L\,i_L^2(0)=0.5*2*20^2=400\,J$$

$$i_y=\frac{40}{10+40}i_x=-16\,e^{-5t}\,A\quad t\geq 0^+$$

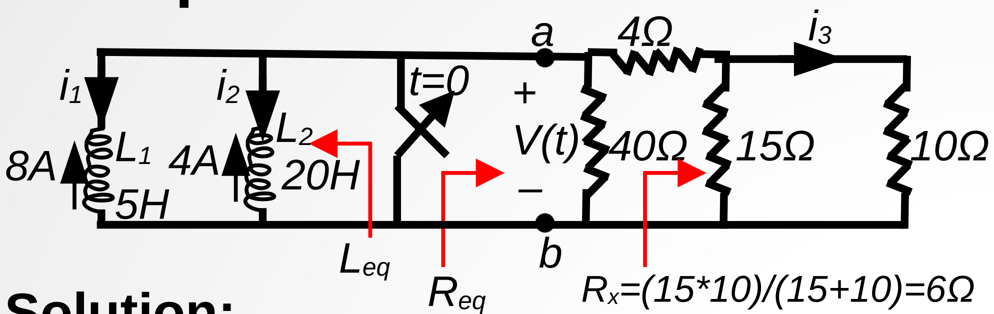
The total energy dissipated in $10\,\Omega$:

$$W_{10\,\Omega}=\int_0^\infty R*i_y^2(t)\,dt=\int_0^\infty 10*16^2 e^{-10t}\,dt=256\,J$$

The percantage of energy disipated in the $10\,\Omega$:

$$\frac{256}{400}*100=64\,\%$$

Example



-The initial currents in inductors L_1 and L_2 have been established by sources that is not shown here.

a) $i_1(t)$, $i_2(t)$, $i_3(t)$ =? for t≥0

b) Calculate the initial energies stored in the inductors.

c) Determine how much energy stored in the inductors as t $\rightarrow \infty$.

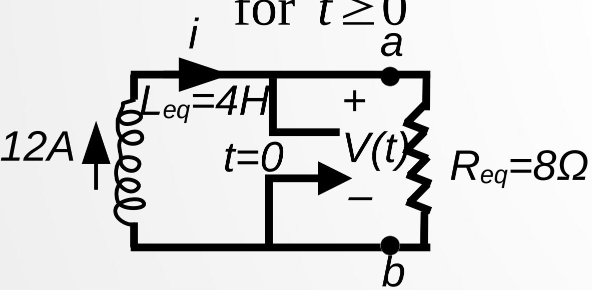
Solution:

$$R_{eq}=40\,\Omega\,//(4\,\Omega+(15\,\Omega\,//\,10\,\Omega))=\frac{40*10}{40+10}=8\,\Omega$$

$$L_{eq}=\frac{20*5}{20+5}=4\,H \qquad i_1(0)=-8\,A \quad i_2(0)=-4\,A$$

$$i(0)=-i_1(0)-i_2(0)=12\,A$$

for $t\geq 0^+$



KVL for the loop $\quad L\frac{di}{dt}+Ri=0$

$$\Rightarrow i(t)=i(0)e^{-\frac{R_{eq}}{L_{eq}}t}=12e^{-\frac{8}{4}t}=12e^{-2t} \quad t\geq 0^+$$

$$V(t)=R_{eq}*i(t)=96e^{-2t}v \quad t\geq 0^+$$

$$i_1(t)=\frac{1}{L_1}\int_0^t V(\tau)d\tau+i_1(0)$$

$$=\frac{1}{5}\int_0^t 96e^{-2\tau}d\tau-8=1.6-9.6e^{-2t}A \quad t\geq 0^+$$

$$i_2(t)=\frac{1}{L_2}\int_0^t V(\tau)d\tau+i_2(0)$$

$$=\frac{1}{20}\int_0^t 96e^{-2\tau}d\tau-4=-1.6-2.4e^{-2t}A \quad t\geq 0^+$$

$$i_3(t)=\frac{V(t)}{10}*\frac{15}{10+15}=5.76e^{-2t}A \quad t\geq 0^+$$

b) The initial energy stored in the inductors:

$$W_{L_1}(0)=\frac{1}{2}L_1 i_1^2(0)=0.5*5*64=160\,J$$

$$W_{L_2}(0)=\frac{1}{2}L_2 i_2^2(0)=0.5*20*16=160\,J$$

$$W_L(0)=W_{L_1}(0)+W_{L_2}(0)=160+160=320\,J$$

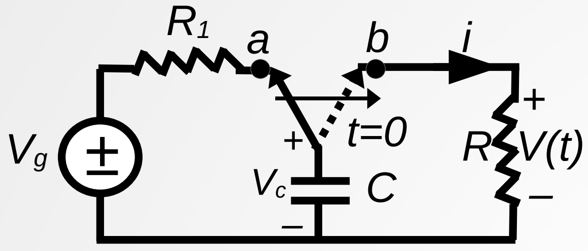
c) As $t\rightarrow\infty$ $\quad i_1(\infty)=1.6\,A$, $\quad i_2(\infty)=-1.6\,A$

$$\Rightarrow \quad W_{L_1}(\infty)=\frac{1}{2}L_1 i_1^2(\infty)=0.5*5*1.6^2=6.4\,J$$

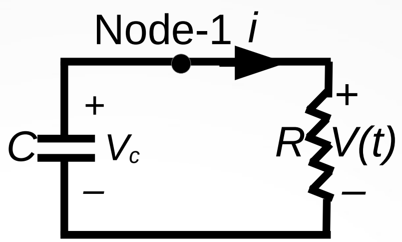
$$W_{L_2}(\infty)=\frac{1}{2}L_2 i_2^2(\infty)=0.5*20*(-1.6)^2=25.6\,J$$

$$W_L(\infty)=W_{L_1}(\infty)+W_{L_2}(\infty)=6.4+25.6=32\,J$$

The Natural Response of an RC Circuit



An RC circuit



The RC circuit after switching (t>0)

KCL for the node-1 $\quad C\frac{dV_c}{dt}+\frac{V_c}{R}=0 \quad \Rightarrow \frac{dV_c}{V_c}=-\frac{1}{RC}dt \quad \Rightarrow \int_{V_c(t_0)}^{V_c(t)}\frac{dx}{x}=-\frac{1}{RC}\int_{t_0}^{t}dy$

$$\Rightarrow \ln\frac{V_c(t)}{V_c(t_0)}=-\frac{1}{RC}(t-t_0) \quad \Rightarrow V_c(t)=V_c(t_0)e^{-\frac{1}{RC}(t-t_0)} \quad \text{for } t\geq t_0$$

$$\text{If } t_0=0 \quad \Rightarrow V_c(t)=V_c(0)e^{-\frac{t}{RC}} \quad \text{for } t\geq 0$$

$$V_c(0^-)=V_c(0^+)=V_c(0)=V_g=V_0, \quad \tau=RC \quad \Rightarrow V_c(t)=V_0e^{-\frac{t}{\tau}} \quad \text{for } t\geq 0$$

$$i(t)=\frac{V_c(t)}{R}=\frac{V_0}{R}e^{-\frac{t}{\tau}} \quad \text{for } t\geq 0^+$$

$$P=V_c(t)*i(t)=\frac{V_0^2}{R}e^{-\frac{2t}{\tau}} \quad \text{for } t\geq 0^+$$

$$W=\int_0^t P(x)dx=\int_0^t\frac{V_0^2}{R}e^{-\frac{2x}{\tau}}dx=\frac{1}{2}CV_0^2\left(1-e^{-\frac{2t}{\tau}}\right) \quad \text{for } t\geq 0$$

Example

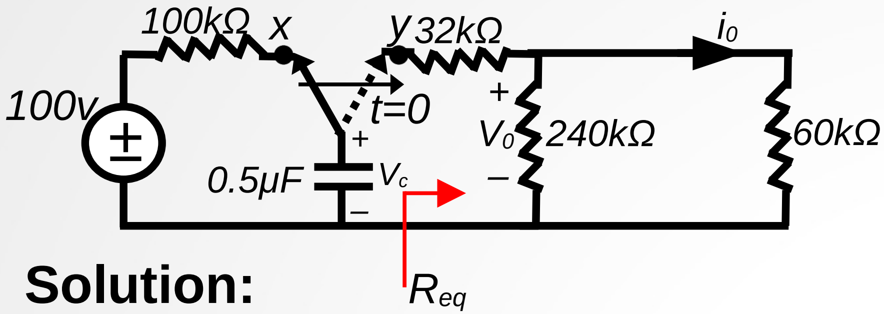
-The switch has been in position *x* for a long time. At time *t=0*, the switch moves instantenously to position *y*.

a) $V_c(t)$ =? for t≥0.

b) $V_0(t)$ =? for $t \geq 0^+$.

c) $i_0(t)$ =? for $t \geq 0^+$.

d) The total energ dissipated in the 60kΩ resistor?

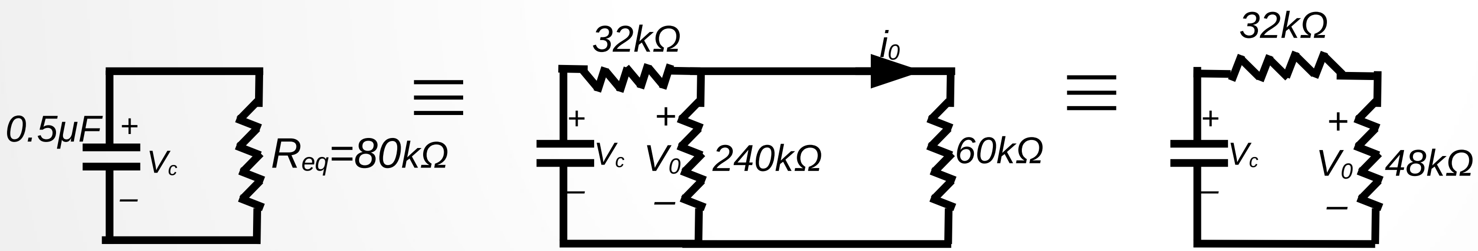


Solution:

$$V_c(0^+) = V_c(0) = V_c(0^-) = 100\,v$$

$$R_{eq} = 32 + \frac{60*240}{60+240}\,k\Omega = 80\,k\Omega \qquad \tau = C*R_{eq} = 0.5*10^{-6}\,80*10^{3} = 40\,msec = 40*10^{-3}\,sec$$

for $t \geq 0$



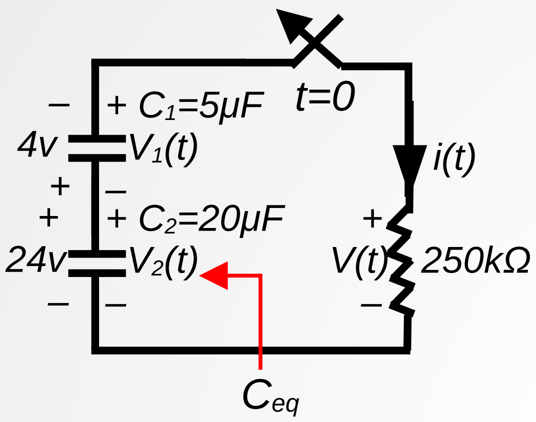
$$V_c(t) = V_c(0)\,e^{-\frac{t}{\tau}} = 100\,e^{-\frac{t}{40 10^{-3}}} = 100\,e^{-25t}\,v \quad \text{for } t \geq 0$$

c) $$i_0(t) = \frac{V_0(t)}{60\,k\Omega} = e^{-25t}\,mA \quad \text{for } t \geq 0^+$$

b) $$V_0(t) = \frac{48}{48+32} * V_c(t) = 60\,e^{-25t}\,v \quad \text{for } t \geq 0^+$$

d) $$W = \int_0^{\infty} i^2 R\,dt = \int_0^{\infty} 60*10^{-3}\,e^{-50t}\,dt = -1.2*10^{-3}\,e^{-50t}\,\Big|_0^{\infty} = 1.2*10^{-3}\,Joules$$

Example



-The initial voltages on capacitors C_1 and C_2 have been established by sources that is not shown here. The switch is closed at *t=0*.

a) $V_1(t)$, $V_2(t)$ =? for t≥0 ,and i(t) =? for $t \geq 0^+$

b) Calculate the initial energies stored in the capacitors C_1 and C_2.

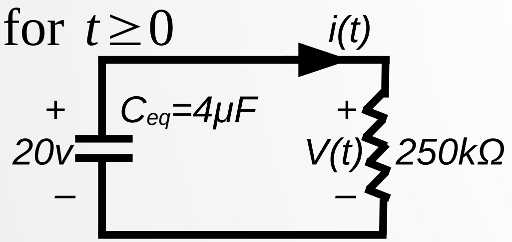
c) Determine how much energy stored in the capacitors as $t \rightarrow \infty$.

d) Show that the total energ delivered to the *250kΩ* resistor is the difference between the result obtained in (b) and (c).

$$V_1(0^-)=V_1(0^+)=-4\,v \qquad V_2(0^-)=V_2(0^+)=24\,v \qquad V(0^+)=V_1(0^+)+V_2(0^+)=-4+24=20\,v$$

$$C_{eq}=\frac{5*20}{5+20}\mu F=4\,\mu F \qquad \tau=C_{eq}*R=4*10^{-6}250*10^{3}=1\,sec$$

for $t \geq 0$



$$V(t)=V(0)e^{-\frac{t}{\tau}}=20\,e^{-t}\,v \qquad \text{for } t\geq 0$$

$$i(t)=\frac{V(t)}{250\,k\Omega}=80\,e^{-t}\,\mu A \qquad \text{for } t\geq 0^+$$

$$V_1(t)=\frac{1}{C_1}\int_0^t i(\tau)\,d\tau+V_1(0)=-\frac{10^6}{5}\int_0^t 80*10^{-6}e^{-x}\,dx-4=16\,e^{-t}-20\,v \qquad t\geq 0$$

$V_2(t)=\frac{1}{C_2}\int_0^t i(\tau)d\tau+V_2(0)=-\frac{10^6}{20}\int_0^t 80*10^{-6}e^{-x}dx+24=4e^{-t}+20v \quad t\geq 0$

b) $W_{C_1}(0)=0.5C_1V_1^2(0)=0.5*5*10^{-6}*4^2=40\mu J.$

$W_{C_2}(0)=0.5C_2V_2^2(0)=0.5*20*10^{-6}*24^2=5760\mu J.$

The total energy stored in the two capacitors is: $W(0)=W_{C_1}(0)+W_{C_2}(0)=40+5760=5800\mu J.$

c) As $t\rightarrow\infty \quad V_1\rightarrow -20 \;,\; V_2\rightarrow 20$

Therefore the energies stored in the two capacitors are:

$W_1(\infty)=0.5*C_1*V_1^2(\infty)=0.5*5*10^{-6}*20^2=1000\mu J.$

$W_2(\infty)=0.5*C_2*V_2^2(\infty)=0.5*20*10^{-6}*20^2=4000\mu J.$

$W(\infty)=W_1(\infty)+W_2(\infty)=1000+4000=5000\mu J.$

d) $W_{250k\Omega}=\int_0^\infty 6400e^{-2t}*10^{-12}*250*10^3dt=1.6*10^{-3}(-0.5*e^{-2t})|_0^\infty=800\mu J.$

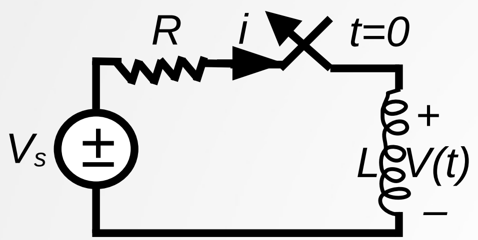
From part b and c

$W(0)=5800\mu J \;, \quad W(\infty)=5000\mu J.$

$\Rightarrow W(0)-W(\infty)=5800-5000=800\mu J=W_{250k\Omega}$

The Step Response of *RL* and *RC* Circuits

The Step Response of an *RL* Circuits



After the switch has been closed, KVL require that:

$$V_s = Ri + L\frac{di}{dt} \quad \Rightarrow \frac{di}{dt} = \frac{-Ri + V_s}{L} = -\frac{R}{L}\left(i - \frac{V_s}{R}\right)$$

$$\Rightarrow di = -\frac{R}{L}\left(i - \frac{V_s}{R}\right)dt \quad \Rightarrow \frac{di}{i - \frac{V_s}{R}} = -\frac{R}{L}dt$$

$$\int_{i(t_0)}^{i(t)} \frac{dx}{x - \frac{V_s}{R}} = -\frac{R}{L}\int_{t_0}^{t} dy$$

$$\Rightarrow \ln\frac{i(t) - V_s/R}{i(t_0) - V_s/R} = -\frac{R}{L}t \quad \Rightarrow i(t) = \frac{V_s}{R} + \left(i(t_0) - \frac{V_s}{R}\right)e^{-\frac{R}{L}(t-t_0)} \quad t \geq t_0$$

If $i(t_0) = 0$ and $t_0 = 0$

$$\Rightarrow i(t) = \frac{V_s}{R} - \frac{V_s}{R}e^{-\frac{R}{L}t} \quad t \geq 0$$

$$V(t) = L\frac{di}{dt} = -L\frac{R}{L}\left(i(t_0) - \frac{V_s}{R}\right)e^{-\frac{R}{L}(t-t_0)} \quad t \geq t_0$$

$$= \left(V_s - Ri(t_0)\right)e^{-\frac{R}{L}(t-t_0)} \quad t \geq t_0$$

If $i(t_0) = 0$ and $t_0 = 0$

$$V(t) = V_s e^{-\frac{R}{L}(t)} \quad t \geq 0$$

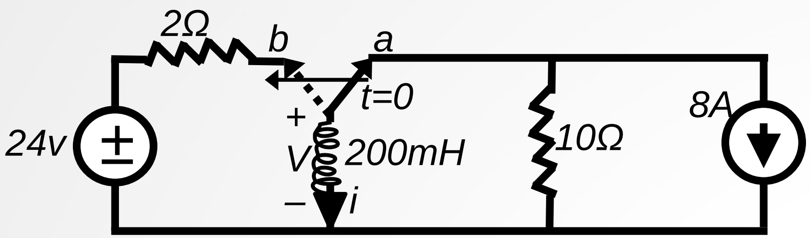
Example

-The switch has been position "*a*" for a long time. At *t=0*, the switch moves from position "*a*" to position "*b*".

a) Find *i(t)* for *t≥0*

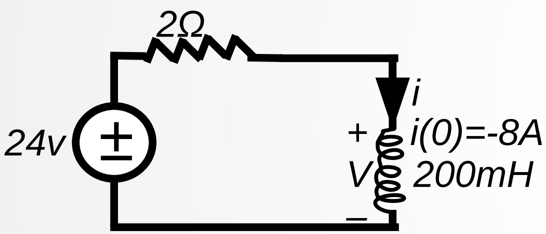
b) What is the initial voltage across the inductor just after the switsh has been moved to position "b"

c) How many milisecond after the switch has moved to position "b" does the inductor voltage equal 24v?



$$i(0) = -8\,A$$

For $t \geq 0$



$$V_s = Ri + L\frac{di}{dt} \quad \Rightarrow i(t) = \frac{V_s}{R} + \left(i(0) - \frac{V_s}{R}\right)e^{-\frac{R}{L}t} \quad t \geq 0$$

$$i(t) = \frac{24}{2} + \left(-8 - \frac{24}{2}\right)e^{-\frac{2}{0.2}t} \quad t \geq 0$$

$$= 12 - 20e^{-10t}\,A \quad t \geq 0$$

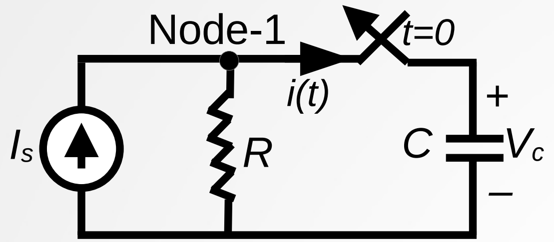
b) $V - 24 + 2i(0) \quad \Rightarrow V = 24 - 2i(0) = 24 + 2*8 = 40\,v$

c) $V = 24 - 2i(t) = 24 - 2\left(12 - 20e^{-10t}\right) = 24$

$$\Rightarrow 12 = 20e^{-10t} \qquad \Rightarrow -10t = \ln\frac{12}{20}$$

$$\Rightarrow t = 0.1 * \ln\frac{20}{12} \qquad \Rightarrow t = 51.08\,ms$$

The Step Response of an RC Circuit



The switch has been closed at *t=0.*

KCL for the node-1

$$C\frac{dV_c}{dt}+\frac{V_c}{R}=I_s \quad \Rightarrow \frac{dV_c}{V_c-I_sR}=-\frac{1}{RC}dt \quad \Rightarrow \int_{V_c(t_0)}^{V_c(t)}\frac{dx}{x-I_sR}=-\frac{1}{RC}\int_{t_0}^{t}dy$$

$$\Rightarrow \ln\frac{V_c(t)-I_sR}{V_c(t_0)-I_sR}=-\frac{1}{RC}(t-t_0) \quad \Rightarrow V_c(t)=RI_s+\left(V_c(t_0)-RI_s\right)e^{-\frac{1}{RC}(t-t_0)} \quad \text{for } t\geq t_0$$

$$\text{If } t_0=0 \quad \Rightarrow V_c(t)=RI_s+\left(V_c(0)-RI_s\right)e^{-\frac{1}{RC}t} \quad \text{for } t\geq 0$$

$$i(t)=I_s-\frac{V_c(t)}{R}=\left(I_s-\frac{V_c(0)}{R}\right)e^{-\frac{1}{RC}t} \quad \text{for } t\geq 0^+$$

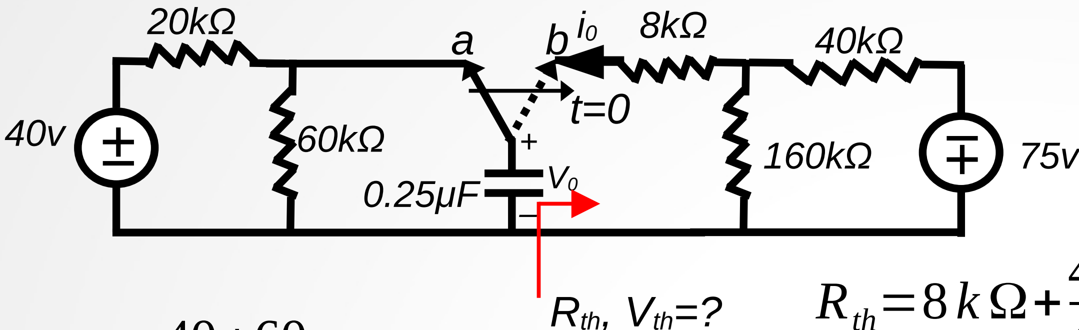
Chapter-7 Response of First Order RL and RC Circuits

Example

-The switch has been in position “a” for a long time. At t=0, the switch moves to position “b”. Find

a) $V_0(t)$ for $t \geq 0$

b) $i_0(t)$ for $t \geq 0^+$

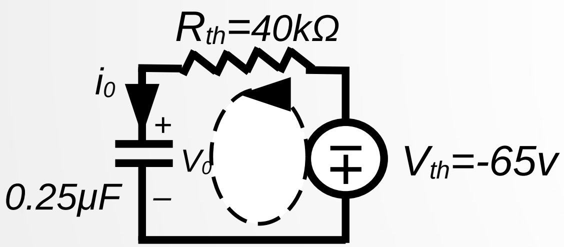


$$R_{th} = 8\,k\Omega + \frac{40*160}{40+160}\,k\Omega = 40\,k\Omega$$

$$V_0(0) = \frac{40*60}{80} = 30\,v$$

$$V_{th} = \frac{-75*160}{40+160} = -60\,v$$

for $t \geq 0$



KVL for the loop

$$R_{eq} i_0 + V_0 + 60 = 0, \quad i_0 = C\frac{dV_0}{dt}$$

$$\Rightarrow R_{eq} C \frac{dV_0}{dt} + V_0 + 60 = 0$$

$$\Rightarrow \frac{dV_0}{V_0+60} = -\frac{1}{R_{eq}C}\,dt$$

$$\Rightarrow \int_{V_0(0)}^{V_0(t)} \frac{dx}{x+60} = -\frac{1}{R_{eq}C}\int_0^t dy \quad \Rightarrow \ln\frac{V_0(t)+60}{V_0(0)+60} = -\frac{1}{R_{eqC}}t$$

$$\Rightarrow V_0(t) = -60 + \left(V_0(0)+60\right)e^{-\frac{1}{R_{eq}C}t} \quad \text{for } t \geq 0$$

$$= -60 + 90\,e^{-100t} \quad \text{for } t \geq 0$$

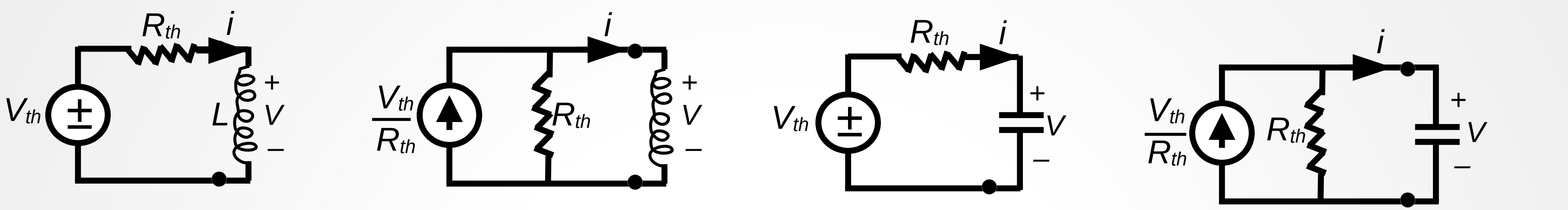
$$i_0(t) = C\frac{dV_0(t)}{dt} = \frac{-60 - V_0(t)}{40\,k\Omega}$$

$$= -2.25 * e^{-100t}\,mA \quad \text{for } t \geq 0^+$$

Chapter-7 Response of First Order RL and RC Circuits

A General Solution for Step and Natural Responses

Four possible first-order circuits



a) An inductor connected To a Thevenin equivalent

b) An inductor connected to a Norton equivalent

c) A capacitor connected to a Thevenin equivalent

d) A capacitor connected to a Norton equivalent

-The differential equation describing any one of the four circuits takes the form:

$$\Rightarrow \frac{dX}{dt} + \frac{X}{\tau} = K \qquad K \text{ is a constant. Can be zero}$$

Because the sources in the circuit are constant voltages and/or currents, the final value of X will be constant.

$$\Rightarrow \text{as t} \rightarrow \infty \quad \frac{dX}{dt} \rightarrow 0 \quad \text{Hence } X_f = K\,\tau \;\; (X_f \text{ is the final value of the variable) } X$$

$$\Rightarrow \frac{dX}{dt} = -\frac{X}{\tau} + K = -\frac{X - X_f}{\tau} \qquad \Rightarrow \frac{dX}{X - X_f} = -\frac{1}{\tau}dt \qquad \Rightarrow \int_{X(t_0)}^{X(t)} \frac{du}{u - X_f} = -\frac{1}{\tau}\int_{t_0}^{t} dv$$

$$\Rightarrow X(t) = X_f + \left[X(t_0) - X_f\right]e^{-\frac{(t - t_0)}{\tau}} \qquad t \geq t_0$$

Chapter-7 Response of First Order RL and RC Circuits

X_f is the final value of X

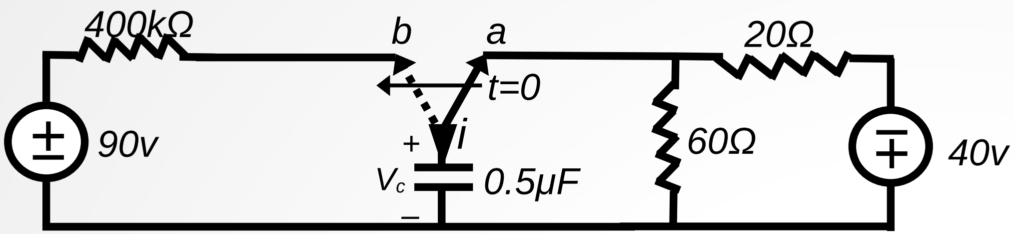
$X(t_0)$ is the initial value of X

τ is the time constant

$\tau = R_{eq} * C_{eq}$ for the RC circuit

$\tau = \frac{L_{eq}}{R_{eq}}$ for the RL circuit

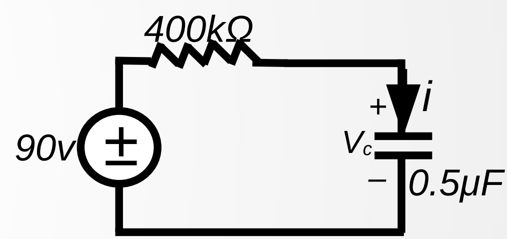
Example



-The switch has been in position “a” for a long time. At t=0, the switch moves to position “b”.

a) What is the initial value of V_c ?
b) What is the final value of V_c ?
c) What is the time constant of the circuit when the switch is in positiob “b” ?
d) $V_c(t)$ for t≥0 ?
e) $i(t)$ for t≥0⁺ ?
f) How long after the switch is in position “b” does the capacitor voltage equal zero?

for $t \geq 0$



a) $V_c(0) = -40 * \frac{60}{60+20} = -30\,v$

b) $V_c(\infty) = V_{cf} = 90\,v$

c) $\tau = R*C = 400*10^3*0.5*10^{-6} = 0.2\,sec$

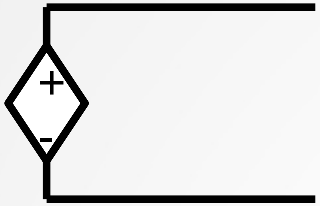
d) $V_c(t) = V_{cf} + [V_c(0) - V_{cf}]\,e^{-\frac{(t-t_0)}{\tau}} = 90 + (-30-90)\,e^{-\frac{t}{0.2}}$

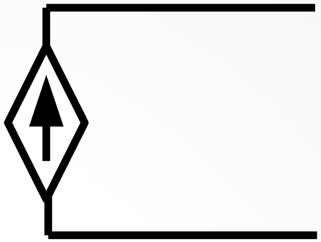
$= 90 - 120\,e^{-5t} \qquad t \geq 0$

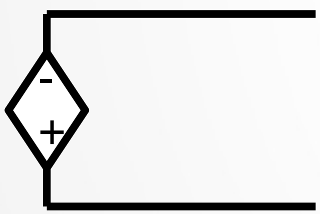
e) $i(t) = C\frac{dV_c}{dt} = \frac{90 - V_c(t)}{400\,k\Omega} = i_f + (i(0) - i_f)\,e^{-\frac{(t-t_0)}{\tau}} = 300\,e^{-5t} \qquad t \geq 0^+$

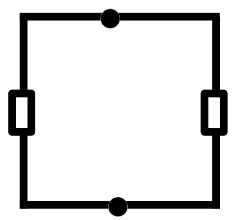
$i_f = i(\infty)$ final value of $i(t)$

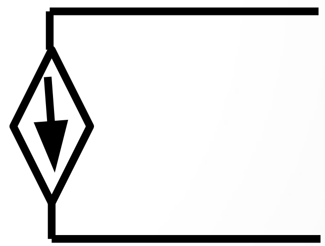
f) $V_c(t) = 0 = 90 - 120\,e^{-5t} \qquad \Rightarrow -5t = \ln\frac{90}{120} \qquad \Rightarrow t = 0.057536\,sec.$

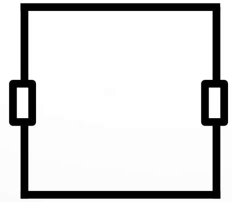


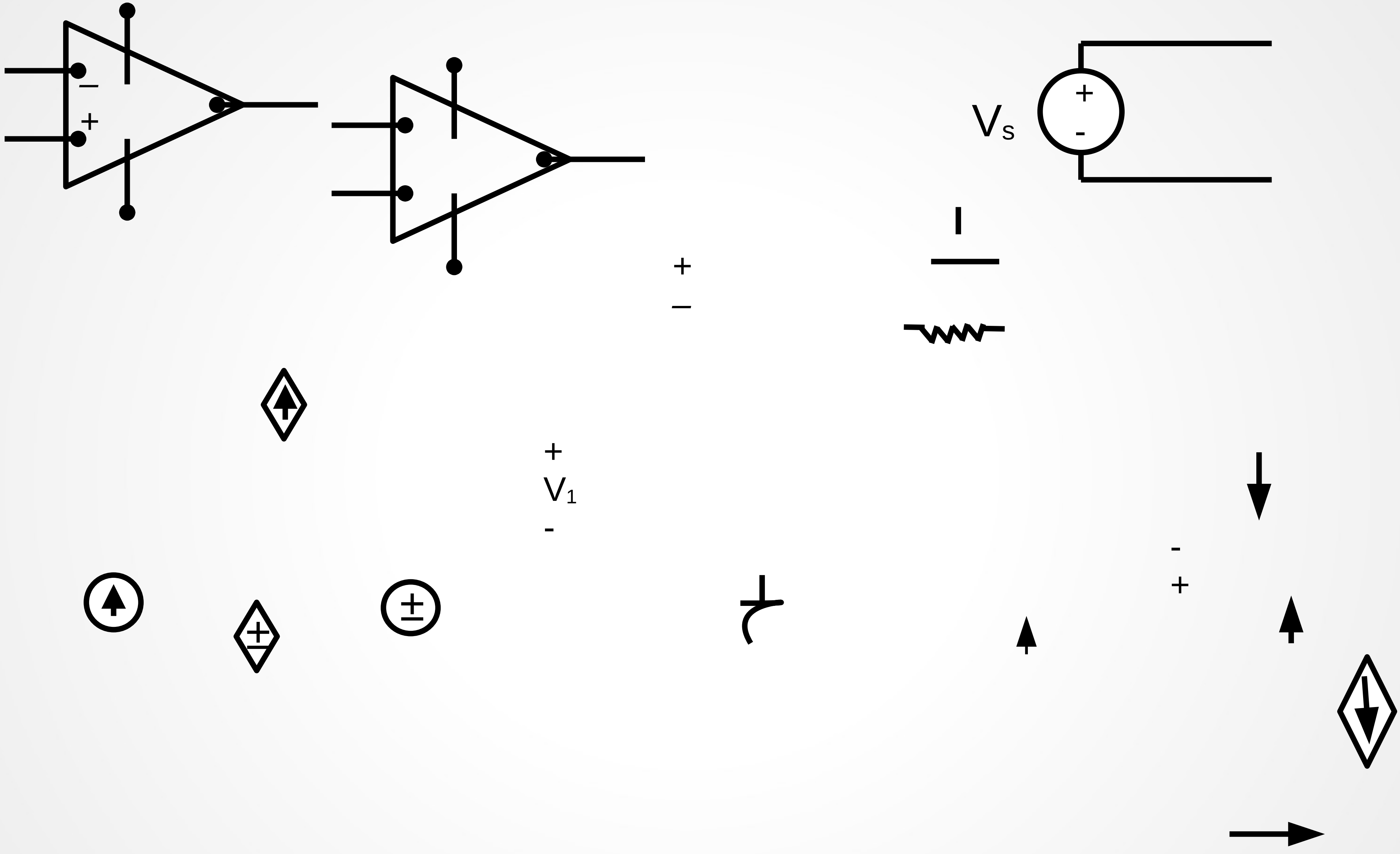












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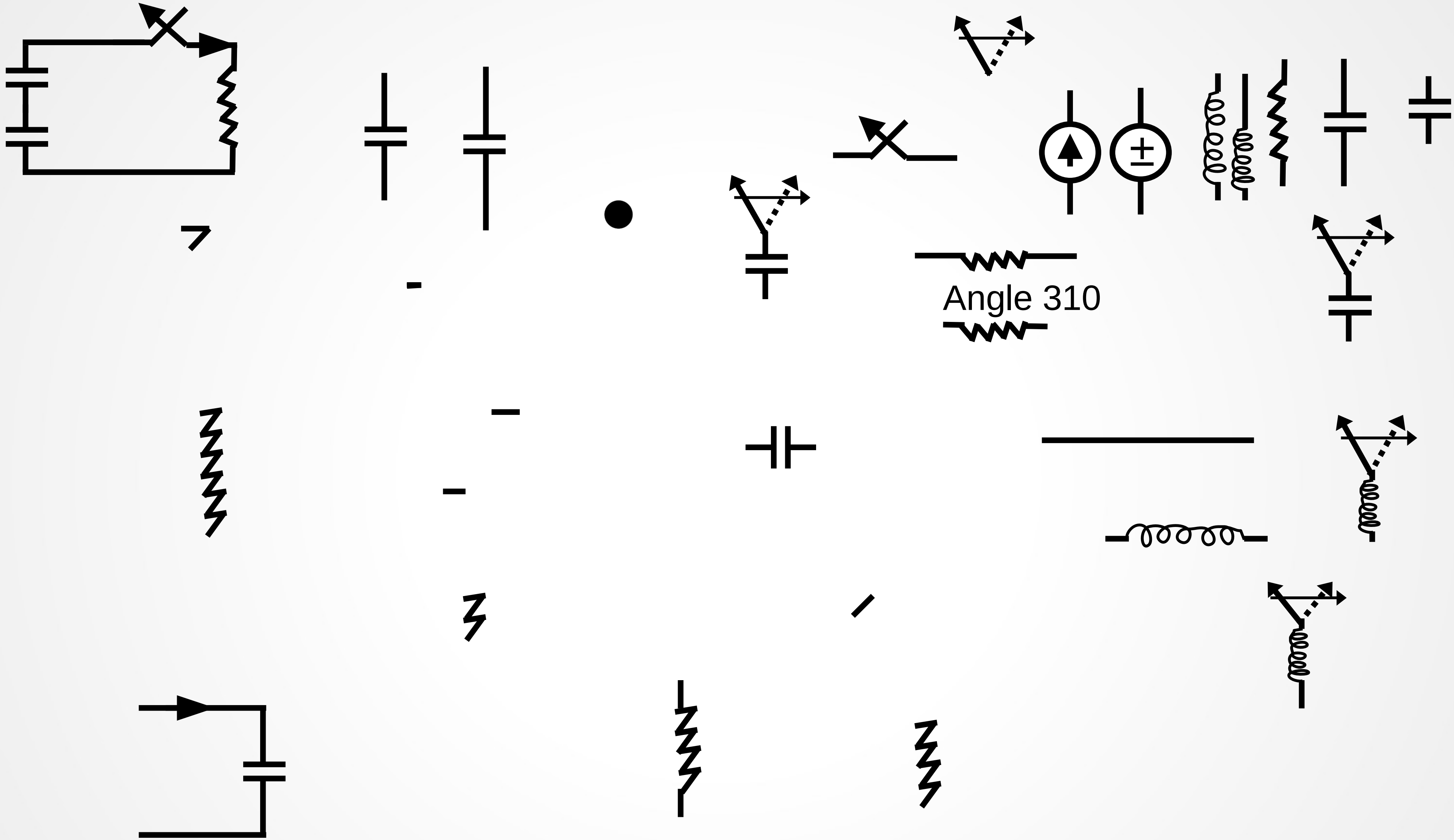
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